WEB DESIGN LAB · LAB 05 HACKATHON

Cortex

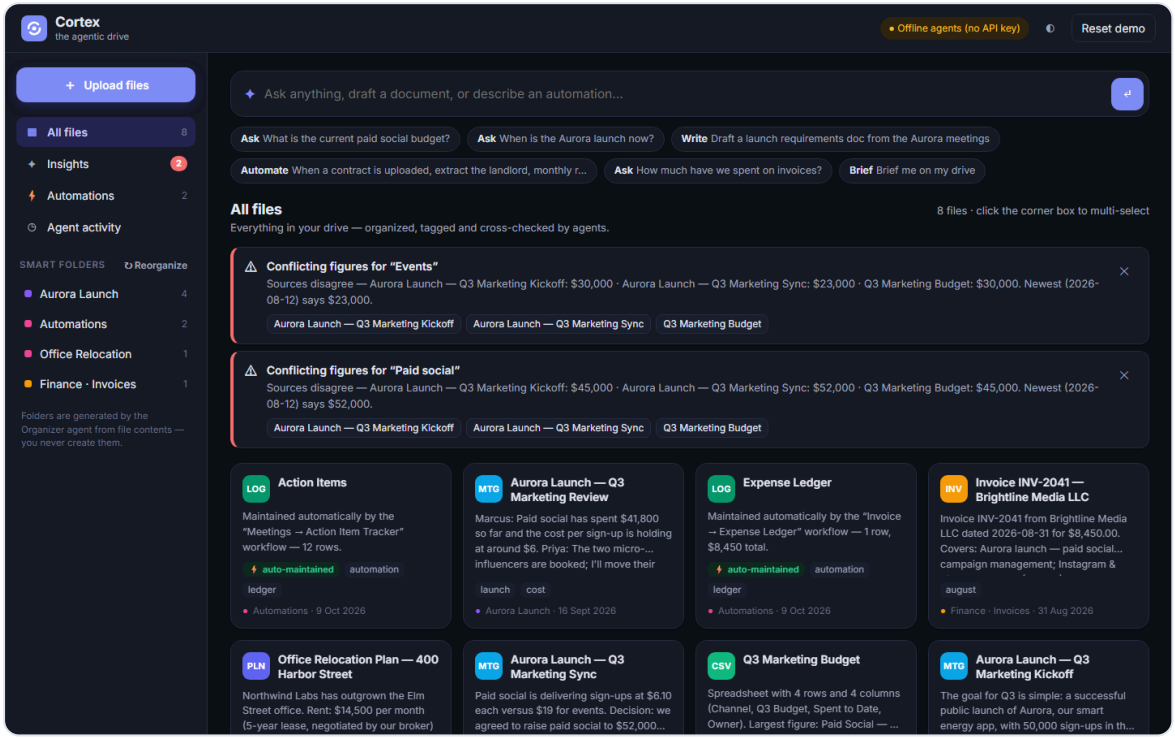
The agentic drive: cloud storage where AI agents read, organize, cross-check and act on your files

Group Member	Roll Number
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GitHub Repository	github.com/ranet5-design/cortex-agentic-drive
Project Page	ranet5-design.github.io/cortex-agentic-drive
Brainstorming Doc	BRAINSTORM.md (in the repository)
Video Pitch	3-minute demo video attached with this submission

1. The Product

> **Problem:** Google Drive and Dropbox are passive filing cabinets. Files go in, nobody reads them, and they quietly contradict each other.
> **Our answer:** a drive that behaves like a diligent colleague who has read every file, remembers what changed, notices when documents disagree, and does the paperwork when new files arrive.



Cortex dashboard: smart folders the agents created, Sentinel conflict alerts, and auto-organized, auto-summarized files.

Core agentic capabilities (all four from the brief are implemented)

1. Proactive Organization

The Librarian titles, summarizes, classifies and tags each upload and extracts people and key figures. The Organizer files it into a smart folder it names itself.

2. Semantic Retrieval

Ask in plain English. Hybrid search (BM25 + AI query expansion) feeds the Analyst, which answers with clickable citations and prefers newer sources.

3. Cross-Document Synthesis

"Draft a requirements doc from the Aurora meetings": the Router picks the sources and the Writer reconciles them into a new *living document*.

4. Automated Workflows

Invoice → Expense Ledger.csv; Meeting notes → Action Items tracker. New rules can be written in plain English and backfill existing files.

Our "aha!" features

- **Sentinel conflict detection:** the lease says **$15,200/month** but the relocation plan budgeted **$14,500**. Cortex flags it the moment the lease is uploaded and highlights the newest source.
- **Self-flagging living documents:** generated docs remember their sources; when related files arrive they get a ↻ **stale** badge and regenerate as v2 in one click.
- **Answers that warn you:** "What is the paid social budget?" returns $52,000 with citations, plus a warning that the budget sheet still says $45,000.
- **Visible agent traces:** every result shows which agent did what, in order, so the AI's work is transparent.

2. Brainstorming Process

Reframing: three lenses

Lens	Passive drive today	Active drive (Cortex)
Memory	Remembers bytes	Remembers *facts*: figures, dates, decisions, and who said them
Attention	Waits for a search	Reacts the moment a file lands: files it, links it, cross-checks it
Agency	Never writes anything	Writes documents, maintains ledgers, refreshes stale docs

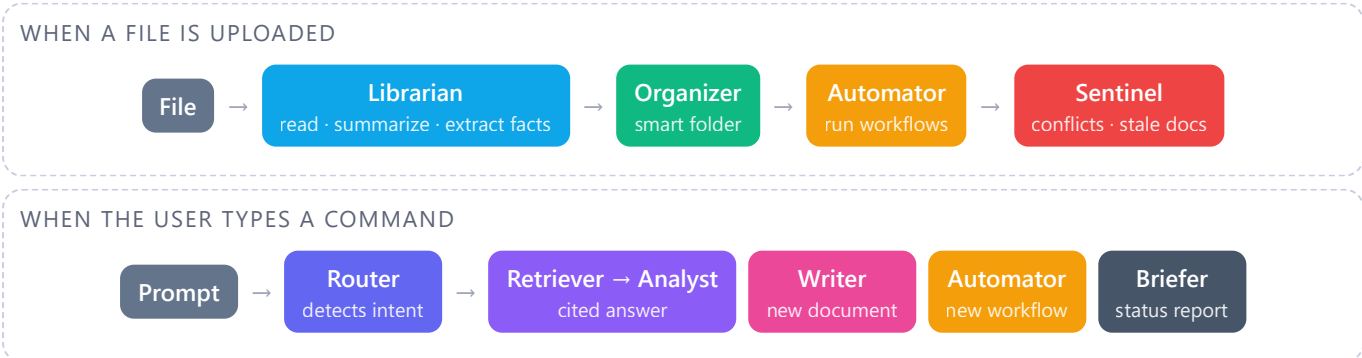
Personas

- **Priya, marketing lead:** lives in meeting transcripts; needs "what did we actually decide?"
- **Dana, ops & finance:** drowns in invoices and contracts; needs numbers extracted and reconciled.
- **Leo, product:** joins mid-project; needs a briefing, not a folder tree.

Idea storm → scoring (impact × "aha" × feasibility)

Idea	Impact	"Aha"	Feasibility	Decision
Cited Q&A over all files	High	Med	High	**Built**
Self-organizing smart folders	High	Med	High	**Built**
Conflict detection across files	Very high	Very high	Med	**Built (signature)**
Living documents that flag themselves stale	High	High	Med	**Built**
Cross-document synthesis, auto source picking	High	High	High	**Built**
Plain-English workflow builder + backfill	High	High	Med	**Built**
Drive briefing ("catch me up")	Med	Med	High	**Built**
Time-travel view of the drive	Med	High	Low	**Future**
Approval flows / deadline reminders	Med	Low	Med	**Future**

Agentic architecture: the intelligent middleware



Each agent has one job, a strict JSON schema and a visible trace. Each also has an offline fallback, so the app never breaks.

3. Agentic User Flows (designed before coding)

Flow A: "The drive caught a mistake"

1. Dana uploads the signed lease.
2. Librarian: contract · Harborview Properties · Monthly Base Rent $15,200.
3. Organizer files it into *Office Relocation*.
4. Sentinel: ⚠ **plan budgets $14,500/month, lease says $15,200** (lease is newer).

Flow B: "Paperwork that does itself"

1. An EventSpace Co invoice lands.
2. *Invoice → Expense Ledger* fires: date, vendor, invoice #, category (Events), total $12,300.
3. A row is appended to **Expense Ledger.csv**, which is itself searchable ("how much have we spent?").

Flow C: "Write me the doc"

1. "Draft a launch requirements doc from the Aurora meetings" (nothing selected).
2. Router picks the 3 meeting transcripts itself; Writer reconciles them (latest figures win, discrepancies listed, action items tabled).
3. Saved as a living document. When customer interview notes arrive later, it is flagged ↻ **stale**, and Refresh produces v2.

Flow D: "Teach the drive a new habit"

1. "When a contract is uploaded, extract the landlord, monthly rent, start date and renewal terms into Contracts.csv"
2. The Automator turns the sentence into a workflow and backfills existing contracts immediately.

Flow E: "Catch me up"

"Brief me on my drive" returns a per-project status, conflicts that need attention, and live ledgers.

4. Technical Execution

Layer	Choice
LLM	Claude Opus 5.5 via the official Anthropic SDK: structured JSON outputs per agent, streaming for long documents, server-side refusal fallback
Retrieval	Passage chunking + BM25 index, plus LLM query expansion (hybrid semantic search) with citations
Backend	Node.js (zero-framework HTTP server), serialized ingest pipeline, JSON-file storage
Frontend	Vanilla HTML/CSS/JS: Drive-style UI, single command bar, drag & drop, live agent traces, dark mode
Resilience	Every agent has a deterministic offline fallback (TF-IDF clustering, extractive answers, regex extraction)

Run it

```
git clone https://github.com/ranet5-design/cortex-agentic-drive.git
cd cortex-agentic-drive && npm install && npm start     # http://localhost:3000
# optional: set ANTHROPIC_API_KEY to power the agents with Claude
```